# 4 Implementation of Discrete Time Systems

## Structures for the Realisation of Discrete Time Systems

Consider a discrete time linear time invariant system characterised by:

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k)$$

$$Y(z) = -\sum_{k=1}^{N} a_k z^{-k} Y(z) + \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$Y(z)\left[1 + \sum_{k=1}^{N} a_k z^{-k}\right] = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)}$$

$$\boxed{H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}}$$

## Structures for FIR System

The FIR system is described by difference equation.

$$y(n) = \sum_{k=0}^{M-1} b_k x(n-k)$$

$$Y(z) = \sum_{k=0}^{M-1} b_k z^{-k} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)}$$

$$\boxed{H(z) = \sum_{k=0}^{M-1} b_k z^{-k}}$$

All poles of FIR are at z = 0. FIR filter is also called all zero filter.

### Unit Sample Response of FIR System

$$h(n) = \begin{cases} b_n, & 0 \le n \le M-1 \\ 0, & \text{otherwise} \end{cases}$$

**1. Direct form structure**

Direct form realisation follows immediately from the non-recursive difference equation.

$y(n) = \sum_{k=0}^{M-1} b_k x(n-k)$ or by the convolution sum

$$y(n) = \sum_{k=0}^{M-1} h(k)\, x(n-k)$$

This structure requires M-1 memory locations for storing the M-1 previous inputs and has a complexity of M multiplications and M-1 additions per output point.

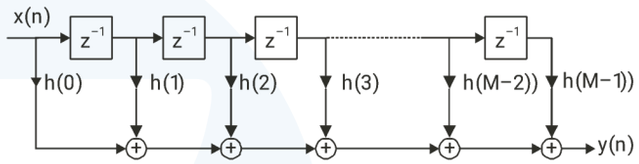



**Fig. 4.1**

### Cascade Form

An alternative to the direct from is suggested by writing H(z) as a product of second order factor as in:

$$H(z) = \prod_{k=1}^{\left[\frac{N}{2}\right]} \left(\beta_{0k} + \beta_{1k} z^{-1} + \beta_{2k} z^{-2}\right)$$

If N is even, one of the coefficients of $\beta_{2k}$ will be zero, corresponding to the fact that for N even, H(z) was an odd number of real roots.

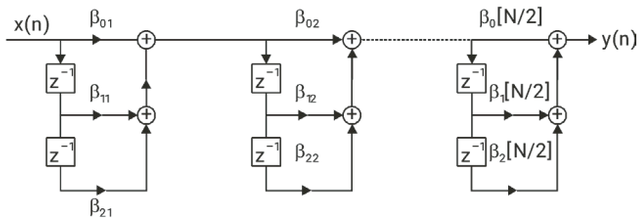



**Fig. 4.2**

**Note:** Cascade form is canonical and also employs N two input adders and N + 1 multipliers for an $N^{th}$ order FIR transfer function.

## Realisation of Linear Phase FIR Systems

The unit sample response for a causal FIR system with linear phase has the property that

$h(n) = h(M - 1 - n)$

The symmetry property of a linear phase FIR filter is used to reduce the multipliers required in the realisation.

$H(z) = Z.T.\ [h(n)]$

$$H(z) = \sum_{n=0}^{M-1} h(n)z^{-n}$$

$$H(z) = \sum_{n=0}^{M-1} h(n)z^{-n} = \sum_{n=0}^{\frac{M}{2}-1} h(n)z^{-n} + \sum_{n=\frac{M}{2}}^{M-1} h(n)z^{-n}$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)z^{-n} + \sum_{n=0}^{\frac{M}{2}-1} h(M-1-n)z^{-(M-1-n)}$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)z^{-n} + \sum_{n=0}^{\frac{M}{2}-1} h(n)z^{-(M-1-n)}$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)[z^{-n} + z^{-(M-1-n)}]$$

For M Even: $$H(z) = \sum_{n=0}^{\frac{M}{2}-1} h(n)\left[z^{-n} + z^{-(M-1-n)}\right]$$

$$Y(z) = H(z)\ X(z)$$

$$= \sum_{n=0}^{\frac{M}{2}-1} h(n)[z^{-n} + z^{-(M-1-n)}]X(z)$$

$$= h(0)\left[1 + z^{-(M-1)}\right]X(z) + h(1)\left[z^{-1} + z^{-(M-2)}\right]X(z)$$

$$+ - - - - + h\left(\frac{M}{2}-1\right)\left[z^{-\left(\frac{M}{2}-1\right)} + z^{\frac{-M}{2}}\right]X(z)$$

$y(n) = h(0)\ [x(n) + x\{n - (M\text{-}1)\}] + h(1)\ [x(n\text{-}1)$
$+ x\{n - (M - 2)\}$

$$+ - - - - + h\left(\frac{M}{2}-1\right)\left[x\left\{n-\left(\frac{M}{2}-1\right)\right\} + x\left(n-\frac{M}{2}\right)\right]$$

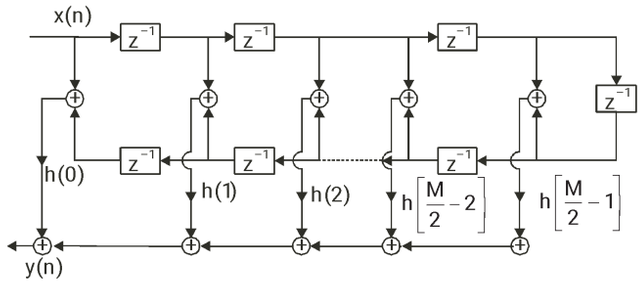



**Fig. 4.3**

For linear phase FIR system, the number of multiplications is reduced from M to M/2 for M even and to $\frac{(M+1)}{2}$ for M odd.

**For M odd:**

$$H(z) = h\left(\frac{M-1}{z}\right)z^{-(M+1)/2} + \sum_{n=0}^{(M-3)/2} h(n)\left[z^{-n} + z^{-(M-1-n)}\right]$$

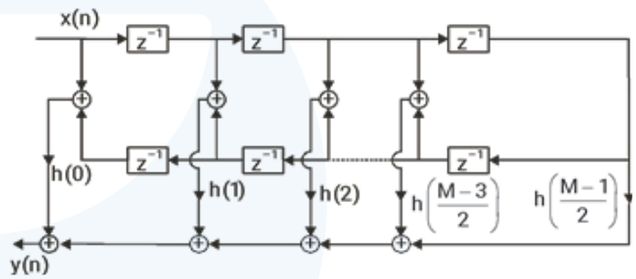



**Fig. 4.4**

**2. Frequency sampling structures**

In this structure parameters that characterise the filter are the values of frequency response instead of impulse response.

To derive the frequency sampling structure, we specify frequency response at the desired set of equally spaced frequencies namely,

$$\omega_k = \frac{2\pi}{M}(k+\alpha),\ k = 0, 1, \ldots \frac{M-1}{2},\ M \text{ odd}$$

$$= 0, 1, \ldots \frac{M}{2} - 1,\ M \text{ even}$$

$$\alpha = 0 \text{ or } \frac{1}{2}$$

$$H(\omega) = \sum_{n=0}^{M-1} h(n) e^{-j\omega n}$$

$$H(k+\alpha) = H\left(\frac{2\pi}{M}(k+\alpha)\right)$$

$$H(\omega)\Big|_{\omega_k = \frac{2\pi}{M}(k+\alpha)}$$

$$= \sum_{n=0}^{M-1} h(n)\, e^{\frac{-j2\pi(k+\alpha)n}{M}},\ k = 0, 1, ----M-1$$

All value of {H(k + α)} are called the frequency samples of H(ω).

If α = 0, {H(k)} corresponds to the M-point DFT of {h(n)}.

$$h(n) = \frac{1}{M}\sum_{k=0}^{M-1} H(k+\alpha)\, e^{j2\pi(k+\alpha)n/M},\ n = 0, 1, ---M-1$$

$$H(z) = \sum_{n=0}^{M-1} h(n) z^{-n}$$

$$H(z) = \sum_{n=0}^{M-1}\left[\frac{1}{M}\sum_{k=0}^{M-1} H(k+\alpha) e^{j2\pi(k+\alpha)n/M}\right] z^{-n}$$

By interchanging the order of the above two summations and performing

$$H(z) = \sum_{n=0}^{M-1} H(k+\alpha)\left[\frac{1}{M}\sum_{k=0}^{M-1}\left(e^{j2\pi(k+\alpha)/M} z^{-1}\right)^n\right]$$

$$H(z) = \frac{1 - z^{-M} e^{j2\pi\alpha}}{M}\sum_{k=0}^{M-1}\frac{H(k+\alpha)}{1 - e^{j2\pi(k+\alpha)/M} z^{-1}}$$

$$H(z) = H_1(z)\, H_2(z)$$

$H_1(z) = \frac{1}{M}\left(1 - z^{-M} e^{j2\pi\alpha}\right)$ → All zero filter or a comb filter.

Its zeros are located at equally spaced points on the unit circle at

$$z_k = e^{j2\pi}(k+\alpha)/M,\ k = 0, 1, ----, M-1$$

$$H_2(z) = \sum_{k=0}^{M-1}\frac{H(K+\alpha)}{1 - e^{j2\pi(k+\alpha)/M} z^{-1}}$$

$H_2(z)$ consists of parallel bank of single pole filter with resonant frequencies:

$$P_k = e^{-j2\pi(k+\alpha)/M},\ k = 0, 1, ----M-1$$

- Poles location are identical to the zero location and they both occur at $\omega_k = 2\pi(k+\alpha)/M$, which are the frequencies at which the desired frequency response is specified.
- When the desired frequency response characterised by the FIR filter is narrow band, most of the gain parameters $\{H(k+\alpha)\}$ are zero.

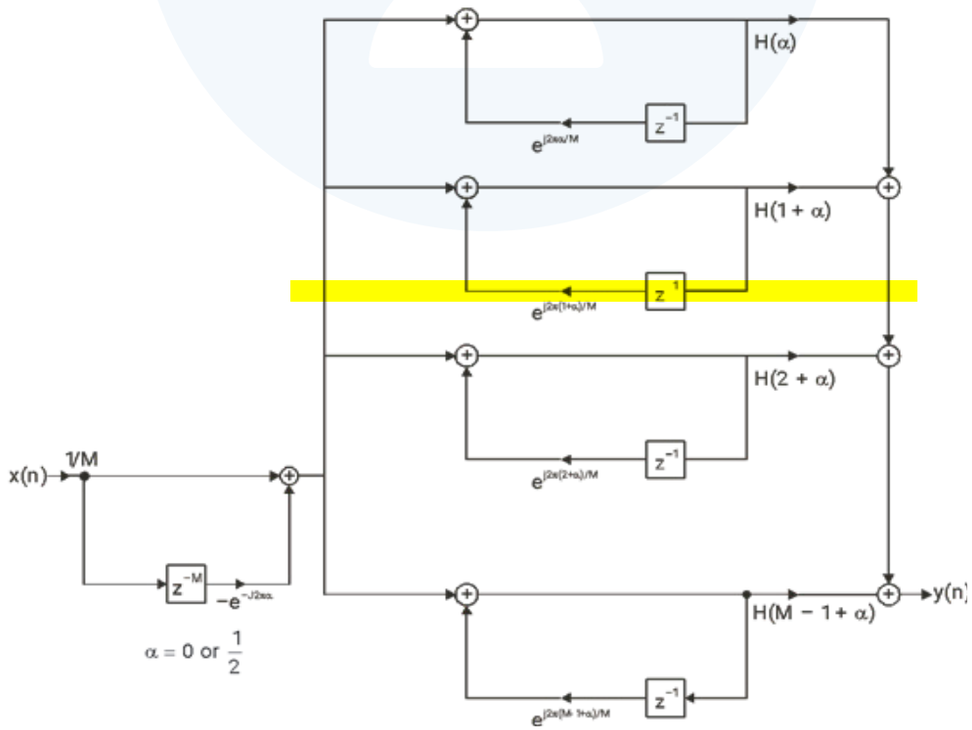



**Fig. 4.5**

The frequency sampling filter structure can be simplified further by exploiting the symmetry in $H(k + \alpha)$.

$H(k) = H^*(M - K)$ for $\alpha = 0$

$$H\left(k+\frac{1}{2}\right) = H\left(M-k-\frac{1}{2}\right) \text{ for } \alpha = \frac{1}{2}$$

## Solved Examples

**1.** Obtain the FIR linear phase realisation and cascade realisation of the system function.

$$H(z) = \left(1+\frac{1}{2}z^{-1}+z^{-2}\right)\left(1+\frac{1}{4}z^{-1}+z^{-2}\right)$$

**Solution:**

**a. Linear phase realisation**

$$H(z) = 1+\frac{3}{4}z^{-1}+\frac{17}{8}z^{-2}+\frac{3}{4}z^{-3}+z^{-4}$$

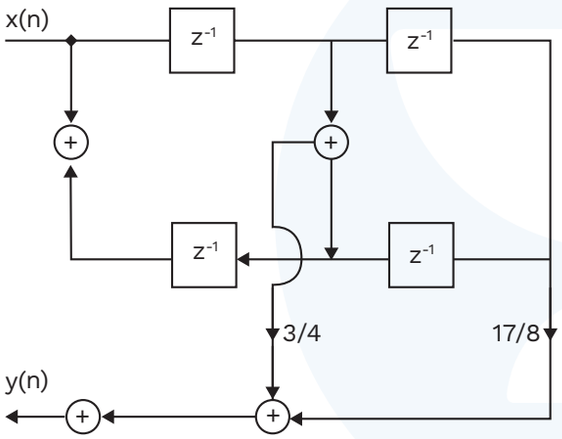



**Fig. 4.6**

**b. Cascade realisation**

$$H(z) = \left(1+\frac{1}{2}z^{-1}+z^{-2}\right)\left(1+\frac{1}{4}z^{-1}+z^{-2}\right)$$

$$= \frac{W(z)}{X(z)} \cdot \frac{Y(z)}{W(z)}$$

$$\frac{W(z)}{X(z)} = 1+\frac{1}{2}z^{-1}+z^{-2}$$

$$w(n) = x(n)+\frac{1}{2}x(n-1)+x(n-2)$$

$$\frac{Y(z)}{W(z)} = 1+\frac{1}{4}z^{-1}+z^{-2}$$

$$y(n) = w(n)+\frac{1}{4}w(n-1)+w(n-2)$$

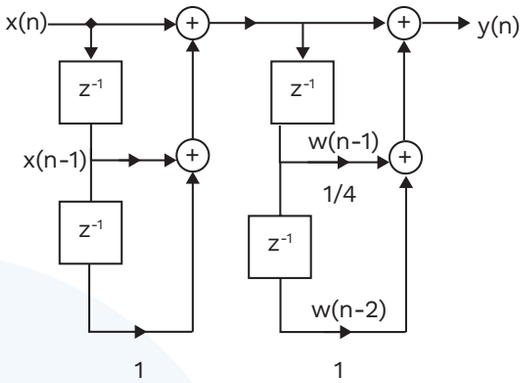



**Fig. 4.7**

**2.** Obtain cascade realisation of the given FIR system function

$$H(z) = 1+\frac{5}{2}z^{-1}+2z^{-2}+2z^{-3}$$

**Solution:**

H(z) can be written in the form

$$H(z) = \left(1+2z^{-1}\right)\left(1+\frac{1}{2}z^{-1}+z^{-2}\right)$$

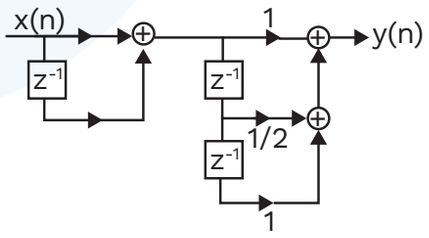



**Fig. 4.8**

**3.** For $H(z) = 1 + 2z^{-1} - z^{-2} + 3z^{-3} + 3z^{-4} - z^{-5} + 2z^{-6} + z^{-7}$, draw the direct form and linear phase FIR implementation. Compare the implementations.

**Solution:**

**a. Direct form structure:** Consider the given system function (length M = 8, order M –1 = 7)

$$H(z) = \frac{Y(z)}{X(z)} = 1 + 2z^{-1} - z^{-2} + 3z^{-3} + 3z^{-4} - z^{-5} + 2z^{-6} + z^{-7}$$

$$Y(z) = X(z) + 2z^{-1}X(z) - z^{-2}X(z) + 3z^{-3}X(z) + 3z^{-4}X(z) - z^{-5}X(z) + 2z^{-6}X(z) + z^{-7}X(z)$$

$$y(n) = x(n) + 2x(n-1) - x(n-2) + 3x(n-3) + 3x(n-4) - x(n-5) + 2x(n-6) + x(n-7)$$

**b. Linear phase structure:** Rewriting this equation, we get:

$$y(n) = [x(n) + x(n-7) + 2[x(n-1) + x(n-6)] - [x(n-2) + x(n-5)] + 3[x(n-3) + x(n-4)]$$

The direct form structure is shown in Fig. 4.9 (a) and the linear phase structure shown in Fig. 4.9 (b). The direct form structure requires M = 8 multipliers and the linear phase structure requires $\frac{M}{2} = 4$ multipliers. Thus, it is clearly seen that the linear phase structure requires 50 per cent fewer multiplications than the direct form.

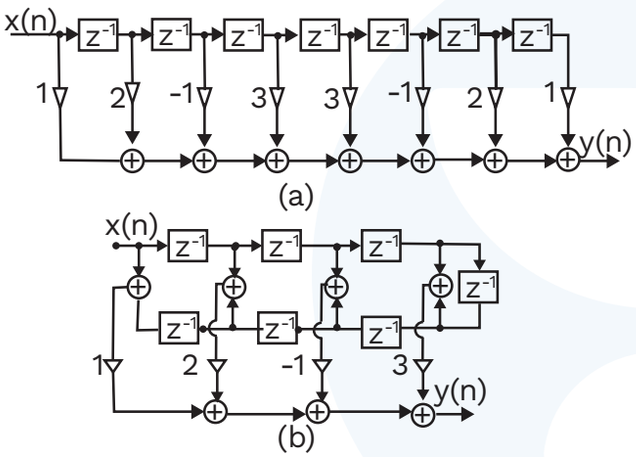



**Fig. 4.9**

**4.** Obtain the cascade form realization of the given FIR filter system function.

$$H(z) = 1 + \frac{6}{5}z^{-1} + \frac{7z^{-2}}{5} + \frac{26z^{-3}}{25} + \frac{1}{5}z^{-4}$$

**Solution:**

H(z) can be written in the form:

$$H(z) = \left(1 + \frac{1}{5}z^{-1} + z^{-2}\right)\left(1 + z^{-1} + \frac{1}{5}z^{-2}\right)$$

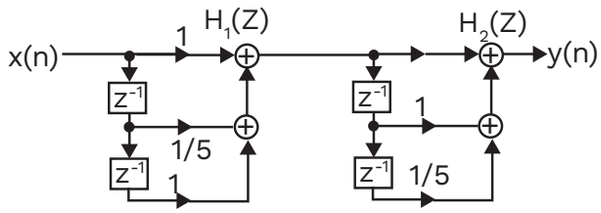



**Fig. 4.10**

**5.** Obtain the linear phase structure for the linear phase FIR system given by:

$$H(z) = 1 + \frac{1}{2}z^{-1} + \frac{1}{3}z^{-2} + \frac{1}{6}z^{-3} + \frac{1}{3}z^{-4} + \frac{1}{2}z^{-5} + z^{-6}$$

**Solution:**

Consider the given system function (length M = 7),

$$H(z) = \frac{Y(z)}{X(z)} = 1 + \frac{1}{2}z^{-1} + \frac{1}{3}z^{-2} + \frac{1}{6}z^{-3} + \frac{1}{3}z^{-4} + \frac{1}{2}z^{-5} + z^{-6}$$

$$Y(z) = \left[1 + z^{-6}\right]X(z) + \frac{1}{2}\left[z^{-1} + z^{-5}\right]X(z) + \frac{1}{3}\left[z^{-2} + z^{-4}\right]X(z) + \frac{1}{6}z^{-3}X(z)$$

$$y(n) = x(n) + x(n-6) + \frac{1}{2}\left[x(n-1) + x(n-5)\right] + \frac{1}{3}\left[x(n-2) + x(n-4)\right] + \frac{1}{6}x(n-3)$$

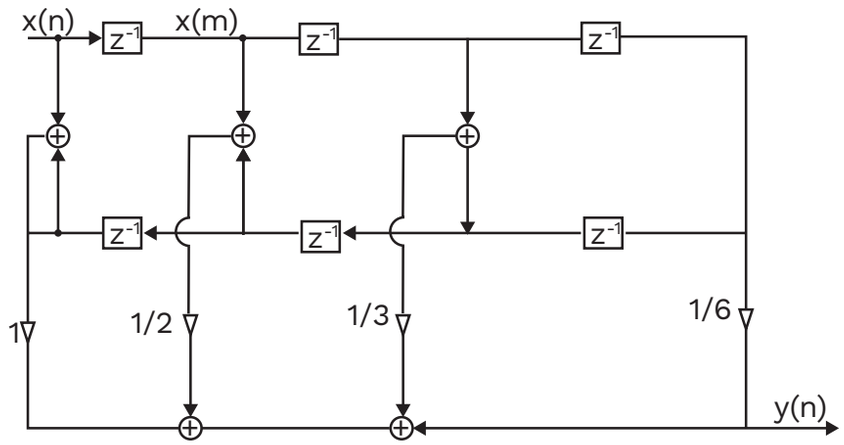



**Fig. 4.11 Linear Phase Frm FIR Structure**

A linear phase structure of the given system function H(z) is shown in Fig. 4.11, which uses only $\frac{(M+1)}{2} = 4$ multipliers.

## Structures of IIR Systems

### Direct Form Structures

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

$$H(z) = H_1(z)\, H_2(z)$$

$$H_1(z) = \sum_{k=0}^{M} b_k z^{-k}$$

$$H_2(z) = \frac{1}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

### Direct Form (I) Realisation

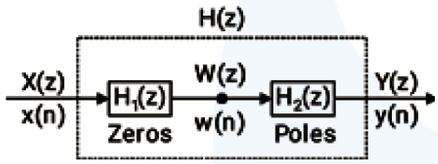



Fig. 4.12

$$H_1(z) = \frac{W(z)}{X(z)} = \sum_{k=0}^{M} b_k z^{-k}$$

$$W(z) = \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$w(n) = \sum_{k=0}^{M} b_k x(n-k)$$

$$\frac{Y(z)}{W(z)} = \frac{1}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

$$Y(z) + \sum_{k=1}^{N} a_k z^{-k} Y(z) = W(z)$$

$$y(n) + \sum_{k=1}^{N} a_k y(n-k) = w(n)$$

$$y(n) = w(n) - \sum_{k=1}^{N} a_k y(n-k)$$

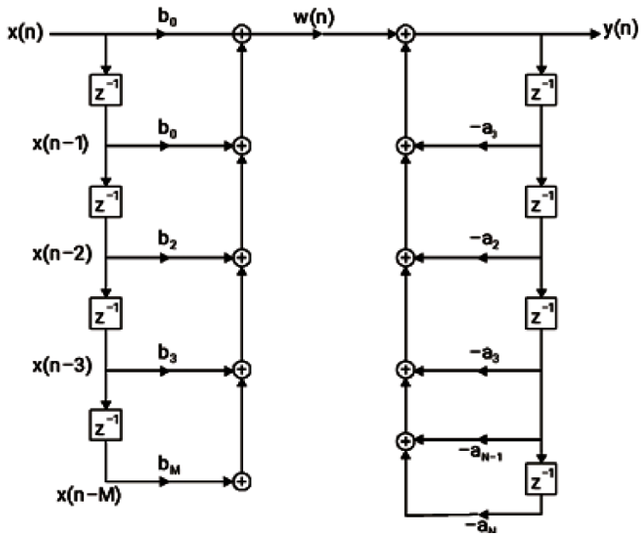



Fig. 4.13

**Note:** Direct Form (I) Realization requires M + N + 1 multiplications, M + N additions and M + N + 1 memory locations.

### Direct Form (II) Realisation

$$H(z) = H_1(z) H_2(z)$$

$$H_1(z) = \frac{1}{1 + \sum_{k=1}^{M} a_k z^{-k}}$$

$$H_2(z) = \sum_{k=0}^{M} b_k z^{-k}$$

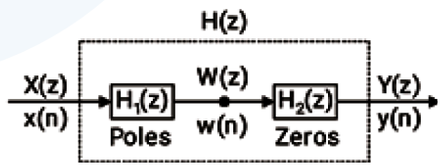



Fig. 4.14

$$H_1(z) = \frac{W(z)}{X(z)} = \frac{1}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

$$W(z) + \sum_{k=1}^{N} a_k z^{-k} W(z) = X(z)$$

$$w(n) + \sum_{k=1}^{N} a_k w(n-k) = x(n)$$

$$\boxed{w(n) = x(n) - \sum_{k=1}^{N} a_k w(n-k)}$$

$$H_2(z) = \frac{Y(z)}{W(z)} = \sum_{k=0}^{M} b_k z^{-k}$$

$$Y(z) = \sum_{k=0}^{M} b_k z^{-k} W(z)$$

$$\boxed{y(n) = \sum_{k=0}^{M} b_k w(n-k)}$$

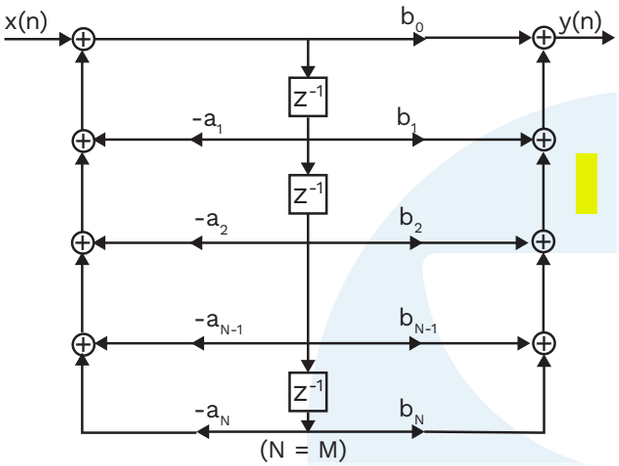



**Fig. 4.15**

**Note:**

1. Direct form (II) structure requires M + N + 1 multiplications, M + N additions, and the maximum of {M, N} memory locations.
2. Direct form (II) realization minimises the number of memory locations and it is said to be canonic.
3. Both structures (Direct form (I) and (II)) are extremely sensitive to parameter quantisation's in general and are not recommended in practical applications.

## Solved Examples

**6.** Determine the direct form I and II for a third order IIR transfer function:

$$H(z) = \frac{0.28z^2 + 0.319z + 0.04}{0.5z^3 + 0.3z^2 + 0.17z - 0.2}$$

**Solution:**

Multiplying the transfer function numerator and denominator by $2z^{-3}$

$$H(z) = \frac{0.56z^{-1} + 0.638z^{-2} + 0.08z^{-3}}{1 + 0.6z^{-1} + 0.34z^{-2} - 0.4z^{-3}}$$

**Direct Form (I) Realization**

$$H(z) = H_1(z) \cdot H_2(z)$$

$$H_1(z) = \frac{W(z)}{X(z)} = 0.56z^{-1} + 0.638z^{-2} + 0.08z^{-3}$$

$$W(z) = \left(0.56z^{-1} + 0.638z^{-2} + 0.08z^{-3}\right) X(z)$$

$$\boxed{w(n) = 0.56x(n-1) + 0.638x(n-2) + 0.08x(n-3)}$$

$$H_2(z) = \frac{Y(z)}{W(z)} = \frac{1}{1 + 0.6z^{-1} + 0.34z^{-2} - 0.4z^{-3}}$$

$$W(z) = \left(1 + 0.6z^{-1} + 0.34z^{-2} - 0.4z^{-3}\right) Y(z)$$

$$w(n) = y(n) + 0.6\, y(n-1) + 0.34\, Y(n-2) - 0.4\, y(n-3)$$

$$y(n) = w(n) - 0.6\, y(n-1) - 0.34\, y(n-2) + 0.4\, y(n-3)$$

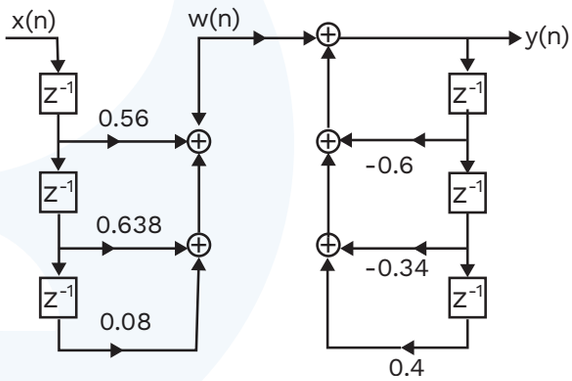



**Fig. 4.16**

**Direct Form (II) Realisation**

$$H(z) = H_1(z) \cdot H_2(z)$$

$$H_1(z) = \frac{W(z)}{X(z)} = \frac{1}{1 + 0.6z^{-1} + 0.34z^{-2} - 0.4z^{-3}}$$

$$X(z) = W(z) + 0.6z^{-1}W(z) + 0.34z^{-2}W(z) - 0.4z^{-3}W(z)$$

$$x(n) = w(n) + 0.6w(n-1) + 0.34\, w(n-2) - 0.4\, w(n-3)$$

$$w(n) = x(n) - 0.6w(n-1) - 0.34\, w(n-2) + 0.4\, w(n-3)$$

$$H_2(z) = \frac{Y(z)}{W(z)} = 0.56z^{-1} + 0.638z^{-2} + 0.08z^{-3}$$

$$Y(z) = 0.56z^{-1}W(z) + 0.638z^{-2}W(z) + 0.08z^{-3}W(z)$$

$$y(n) = 0.56\, w(n-1) + 0.638\, w(n-2) + 0.08\, w(n-3)$$

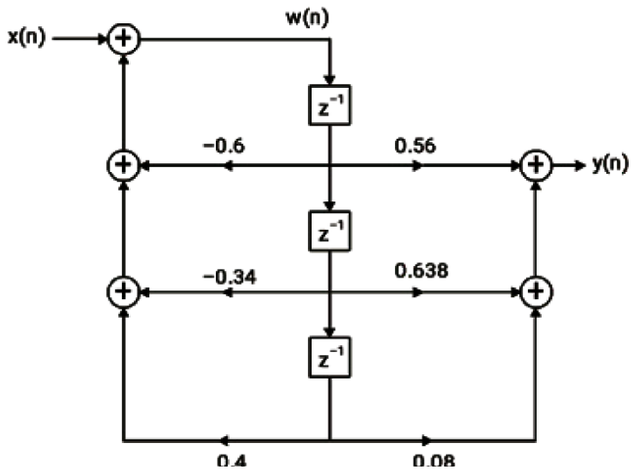



Fig. 4.17

**Signal Flow Graphs and Transposed Structure**

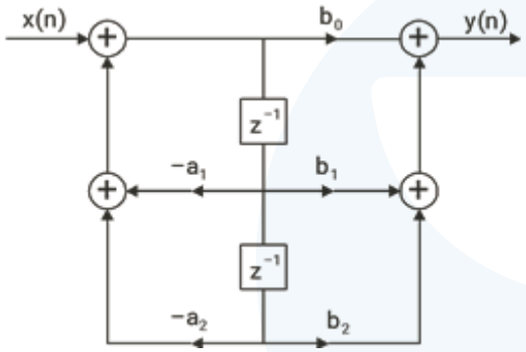



Fig. 4.18

**Signal Flow Graph of above System**

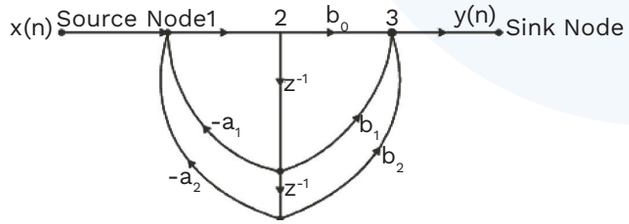



Fig. 4.19

We observe that the signal flow graph contains the same basic information as the block diagram realization of the system.

- The theory of linear signal flow graphs provides a variety of manipulative procedures for transforming signal flow graph into difference forms while leaving the transmission between input and output unchanged.
- One of these procedures is called flow graph reversal theorem or transposition.
- This theorem simply states that if we reverse the direction of all branch transmittances and interchange the input and output in the flow graph, the system function remains unchanged. The resulting structure is called a transposed structure or a transposed form.

## Signal Flow Graph of Transposed Structure and Its Realisation

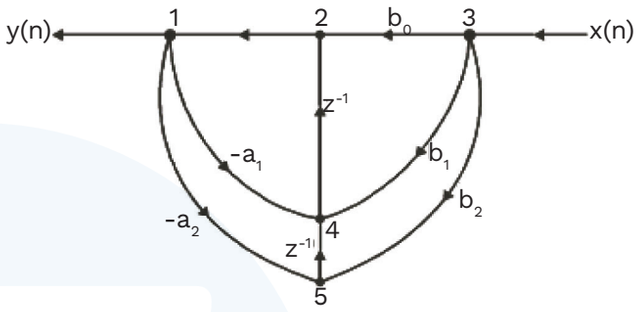



Fig. 4.20

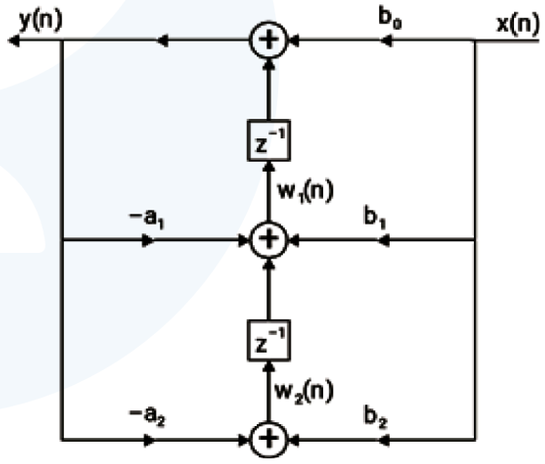



Fig. 4.21

$$y(n) = b_0x(n) + w_1(n-1)$$

$$w_1(n) = b_1x(n) - a_1y(n) + w_2(n-1)$$

$$w_2(n) = b_2x(n) - a_2y(n)$$

**Cascade Form Structures**

$$H(z) = \frac{\prod_{k=1}^{M_1}\left(1-g_kz^{-1}\right)\prod_{k=1}^{M_2}\left(1-h_kz^{-1}\right)\left(1-h_k^*z^{-1}\right)}{\prod_{k=1}^{N_1}\left(1-c_kz^{-1}\right)\prod_{k=1}^{N_2}\left(1-d_kz^{-1}\right)\left(1-d_k^*z^{-1}\right)}$$

Where, $M = M_1 + 2M_2$

$N = N_1 + 2N_2$

It is important to implement the cascade realization using minimum of storage and hardware realization are often implemented by time sharing or multiplexing a single second order section.

- In term of a cascade form based on the equation

$$H(z) = A \prod_{k=1}^{[(N+1)/2]} \frac{1+\beta_{1k}z^{-1}+\beta_{2k}z^{-2}}{1-\alpha_{1k}z^{-1}-\alpha_{2k}z^{-2}}$$

where [(N + 1)/2] means the largest integer contained in (N + 1)/2

- Each of second order subsystem with system function of the above form can be realised in direct form (II).
- If there are an odd number of real zeros, $\beta_{2k} = 0$.
- If there are odd number of real poles, $\alpha_{2k} = 0$.

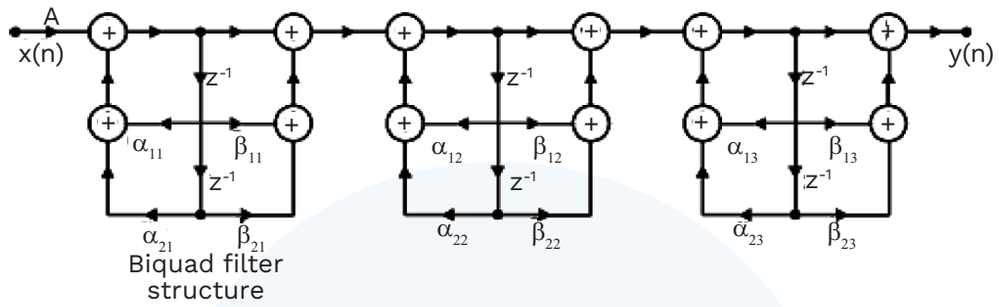



**Fig. 4.22**

### Parallel Form Structure

$$H(z) = C + \sum_{k=1}^{N} \frac{A_k}{1-P_k z^{-2}} \quad N \geq M$$

Where, $\{A_k\} \Rightarrow$ Coefficients (residues)

$$C = \frac{b_N}{a_N}$$

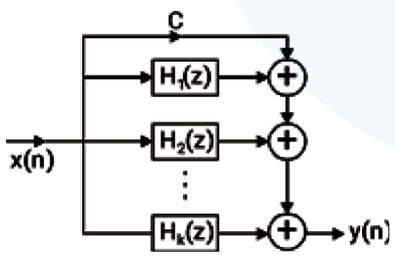



**Fig. 4.23**

$$H_k(z) = \frac{b_{k0}+b_{k1}z^{-1}}{1+a_{k1}z^{-1}+a_{k2}z^{-2}}$$

Where the coefficient $\{b_{ki}\}$ and $\{a_{ki}\}$ are real valued system parameters. The overall function can now be expressed as

$$H(z) = C + \sum_{k=1}^{[(N+1)/2]} H_k(z)$$

Where [(N + 1)/2] means largest integer contained in (N + 1)/2.

Each Biquad system $H_k(z)$ can be implemented in direct form II.

## Structure of Second Order Section in a Parallel IIR System Realisation

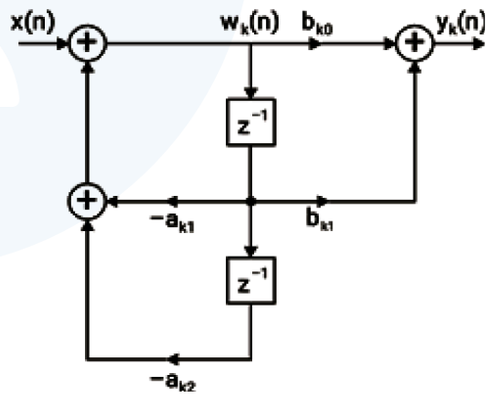



**Fig. 4.24**

$$w_k(n) = -a_{k1}w_k(n-1) - a_{k2}w_k(n-2) + x(n)$$

$$y_k(n) = b_{k0}w_k(n) + b_{k1}w_k(n-1)$$

$$y(n) = cx(n) + \sum_{k=1}^{[(n-1)/2]} y_k(n)$$

**All pass filters:** An all pass filter has a frequency response with a constant magnitude.

$$\left|H_{ap}(e^{j\omega})\right| = 1 \; \forall\omega$$

If the system function of an all pass filter is a rational function of z, it has the form:

$$H_{ap}(z) = \prod_{k=1}^{P} \frac{z^{-1} - a_k^*}{1 - a_k z^{-1}}$$

If h(n) is real valued, the complex roots occurs in conjugate pairs, and these pairs may be combined to form second order factors with real coefficients.

$$H_{ap}(z) = \prod_{k=1}^{[N+1/2]} \frac{\alpha_k + \beta_k z^{-1} + z^{-2}}{1 + \beta_k z^{-1} + \alpha_k z^{-2}}$$

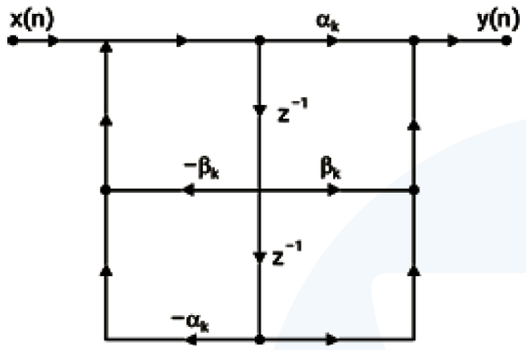



**Fig. 4.25**

## Solved Examples

**7.** Draw the structures of cascade and parallel realisations of:

$$H(z) = \frac{(1 - z^{-1})^3}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{8}z^{-1}\right)}$$

**Solution:**

**Cascade realisation:** The given transfer H(z) can be written in the form:

$$H(z) = \frac{(1 - z^{-1})}{\left(1 - \frac{1}{2}z^{-1}\right)} \frac{(1 - z^{-1})}{\left(1 - \frac{1}{8}z^{-1}\right)} (1 - z^{-1})$$

$$= H_1(z) \cdot H_2(z) \cdot H_3(z)$$

This function is realised using cascade structure as shown in Fig. 4.26 (a).

Parallel realisation: The transfer function H(z) can be written as:

$$H(z) = \frac{(1 - z^{-1})^3}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{8}z^{-1}\right)} = \frac{(z - 1)^3}{z\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)}$$

For finding the parallel realization, the partial fraction expansion of H(z)/z is determined as:

$$F(z) = \frac{H(z)}{z} = \frac{(z - 1)^3}{z^2\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)}$$

$$= \frac{A_1}{z^2} + \frac{A_2}{z} + \frac{A_3}{z - \frac{1}{2}} + \frac{A_4}{z - \frac{1}{8}}$$

Where, $A_1$

$$= \left.\frac{(z - 1)^3}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)}\right|_{z=0} = -16, \; A_3 = \left.\frac{(z - 1)^3}{z^2\left(z - \frac{1}{8}\right)}\right|_{z=\frac{1}{4}} = -\frac{4}{3}$$

$$A_2 = \frac{d}{dz}\left[\frac{(z - 1)^3}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{8}\right)}\right]_{z=0} = -112,$$

$$A_4 = \left.\frac{(z - 1)^3}{z^2\left(z - \frac{1}{2}\right)}\right|_{z=\frac{1}{8}} = -\frac{343}{3}$$

Thus,

$$H(z) = -112 - 16z^{-1} - \frac{4}{3}\frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)} + \frac{343}{3} + \frac{1}{\left(1 - \frac{1}{8}z^{-1}\right)}$$

This function is realised in parallel structure as shown in Fig. 4.26 (b).

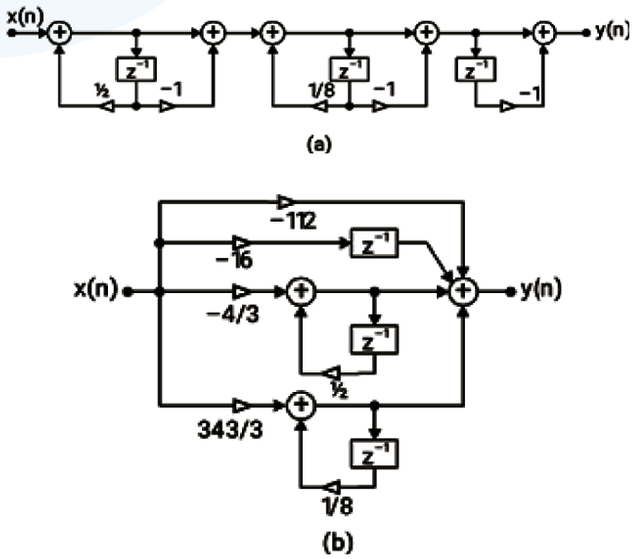



**Fig. 4.26**

**8.** Draw the cascade and parallel realization structure for the system described by the system function:

$$H(z) = \frac{5\left(1-\frac{1}{4}z^{-1}\right)\left(1-\frac{2}{3}z^{-1}\right)\left(1+2z^{-1}\right)}{\left(1-\frac{3}{4}z^{-1}\right)\left(1-\frac{1}{8}z^{-1}\right)\left[1-\left(\frac{1}{2}+j\frac{1}{2}\right)z^{-1}\right]\left[1-\left(\frac{1}{2}-j\frac{1}{2}\right)z^{-1}\right]}$$

**Solution:**

**Cascade realisation:**

$$H_1(z) = \frac{1-\frac{2}{3}z^{-1}}{1-\frac{7}{8}z^{-1}+\frac{3}{32}z^{-2}}, H_2(z) = \frac{1+\frac{7}{4}z^{-1}-\frac{1}{2}z^{-2}}{1-z^{-1}+\frac{1}{2}z^{-2}}$$

Hence, $H(z) = 5H_1(z)H_2(z)$

The difference equations for $H_1(z)$ and $H_2(z)$ are:

$$y_1(n) = \frac{7}{8}y_1(n-1)-\frac{3}{32}y_1(n-2)+x_1(n)-\frac{2}{3}x_1(n-1)$$

$$y_2(n) = y_2(n-1)-\frac{1}{2}y_2(n-2)+x_2(n)+\frac{7}{4}x_2(n-1)-\frac{1}{2}x_2(n-2)$$

These functions are realized in cascade form as shown in below the Fig. (a)

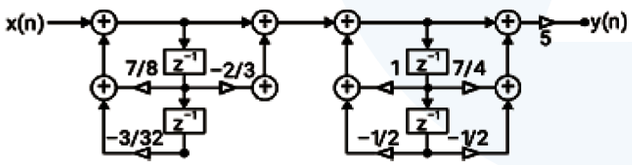



**Fig. 4.27**

**Parallel realisation:** To obtain parallel form realization, H(z) must be expanded in partial fractions. Thus we have:

$$H(z) = \frac{A_1}{1-\frac{3}{4}z^{-1}}+\frac{A_2}{1-\frac{1}{8}z^{-1}}+\frac{A_3}{1-\left(\frac{1}{2}+j\frac{1}{2}\right)z^{-1}}+\frac{A_3^*}{1-\left(\frac{1}{2}-j\frac{1}{2}\right)z^{-1}}$$

Upon solving, we find $A_1 = 2.933$, $A_2 = -2.947$, $A_3 = 2.507 - j10.45$ and $A_3^* = 2.507 + j10.45$

Therefore, H(z)

$$= \frac{2.933}{\left(1-\frac{3}{4}z^{-1}\right)}-\frac{2.947}{\left(1-\frac{1}{8}z^{-1}\right)}+\frac{2.507-j10.45}{1-\frac{1}{2}(1+j)z^{-1}}+\frac{2.507+j10.45}{1-\frac{1}{2}(1-j)z^{-1}}$$

$$= \frac{-0.02+1.846z^{-1}}{1-\frac{7}{8}z^{-1}+\frac{3}{32}z^{-2}}+\frac{5.02+7.743z^{-1}}{1-z^{-1}+\frac{1}{2}z^{-2}}$$

The parallel form realization is illustrated in Fig. (b)

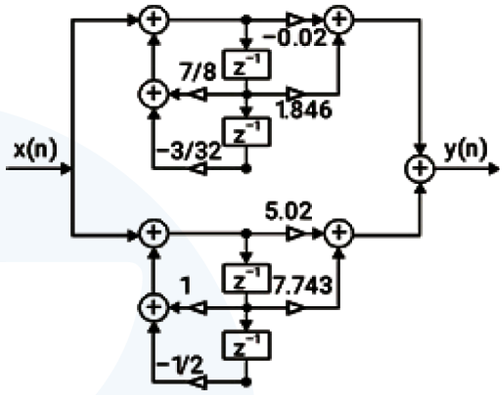



**Fig. 4.28**

**9.** Obtain the cascade and parallel realisations for the system function given by:

$$H(z) = \frac{1+\frac{1}{4}z^{-1}}{\left(1+\frac{1}{2}z^{-1}\right)\left(1+\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}\right)}$$

**Solution:**

**Cascade realisation:**

$H(z) = H_1(z)H_2(z)$

Where, $H_1(z) = \frac{1+\frac{1}{4}z^{-1}}{1+\frac{1}{2}z^{-1}}$ and $H_2(z) = \frac{1}{1+\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$

The cascade realization structure for this system function is shown in Fig. (a)

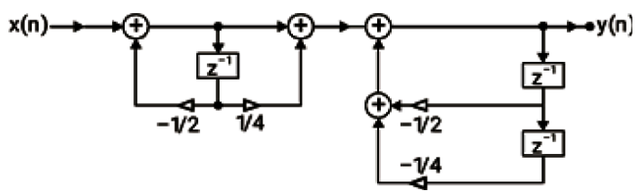



**Fig. 4.29**

**Parallel realisation:** To obtain the parallel realization, we make a partial fraction expansion of H(z) resulting in:

$$H(z) = \frac{1+\frac{1}{4}z^{-1}}{\left(1+\frac{1}{2}z^{-1}\right)\left(1+\frac{1}{2}z^{-1}+\frac{1}{4}z^{-1}\right)}$$

$$= \frac{A_1}{\left(1+\frac{1}{2}z^{-1}\right)} + \frac{A_2z^{-1}+A_3}{1+\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

Multiplying and equating the coefficients of negative powers of z and the solving, we obtain,

$$A_1 = \frac{1}{2},\ A_2 = -\frac{1}{4} \text{ and } A_3 = \frac{1}{2}$$

$$\text{Therefore, } H(z) = \frac{\frac{1}{2}}{1+\frac{1}{2}z^{-1}} + \frac{\frac{1}{2}-\frac{1}{4}z^{-1}}{1+\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

The parallel realization structure is shown in Fig. (b)

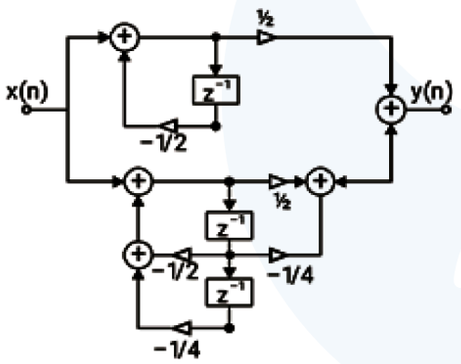



**Fig. 4.30**

**10.** Develop cascade and parallel realization structures for:

$$H(z) = \frac{\frac{z}{6}+\frac{5}{24}+\frac{5}{24}z^{-1}+\frac{1}{24}z^{-2}}{1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

**Solution:**

Cascade realisation: The transfer function can be rearranged as:

$$H(z) = \frac{\frac{z}{6}\left(1+\frac{5}{4}z^{-1}+\frac{5}{4}z^{-2}+\frac{1}{4}z^{-3}\right)}{1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

$$= \frac{\frac{z}{6}\left(1+\frac{1}{4}z^{-1}\right)(1+z^{-1}+z^{-2})}{1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

$$= \frac{1}{6}z(1+z^{-1}+z^{-2})\frac{1+\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

The above function is in the form of $H(z) = \frac{1}{6}H_1(z)\cdot H_2(z)\cdot H_3(z)$.

There are two ways of implementing this system, depending on the combination of numerator term with the denominator term.

The cascade realization structure is shown in Fig. (a)

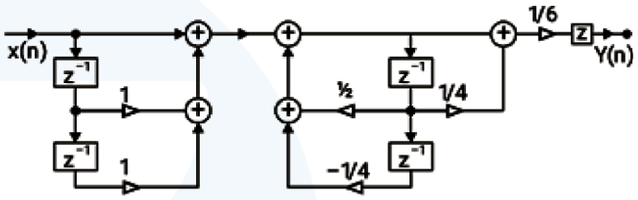



**Fig. 4.31**

**Parallel realisation:**

$$H(z) = \frac{\frac{z}{6}+\frac{5}{24}+\frac{5}{24}z^{-1}+\frac{1}{24}z^{-2}}{1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

$$= \frac{\frac{z}{6}\left(1+\frac{5}{4}z^{-1}+\frac{5}{4}z^{-2}+\frac{1}{4}z^{-3}\right)}{1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2}}$$

Dividing the numerator by the denominator, we obtain:

$$\begin{array}{r|l} & z^{-1}+7 \\ \hline \frac{1}{4}z^{-2}-\frac{1}{2}z^{-1}+1 & \frac{1}{4}z^{-3}+\frac{5}{4}z^{-2}+\frac{5}{4}z^{-1}+1 \\ & \frac{1}{4}z^{-3}-\frac{1}{2}z^{-2}+z^{-1} \\ \hline & \frac{7}{4}z^{-2}+\frac{1}{4}z^{-1}+1 \\ & \frac{7}{4}z^{-2}+\frac{7}{2}z^{-1}+7 \\ \hline & \frac{15}{4}z^{-1}-6 \end{array}$$

Therefore,

$$H(z) = \frac{z}{6}\left[7 + z^{-1} + \frac{-6 + \frac{15}{4}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}\right]$$

$$= z\left[\frac{7}{6} + \frac{1}{6}z^{-1} + \frac{-1 + \frac{5}{8}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}\right]$$

$$H(z) = z[H_1(z) + H_2(z)]$$

Where, $H_1(z) = \frac{7}{6} + \frac{1}{6}z^{-1}$ and $H_2(z) = \frac{-1 + \frac{5}{8}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$

The parallel realization structure is shown in Fig. (b)

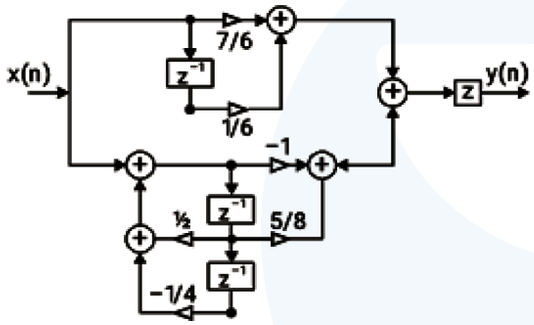



**Fig. 4.32**

**11.** Obtain (a) direct form I and (b) direct form II (canonical realization of the system function).

$$H(z) = \frac{1 + 2z^{-1} - z^{-2}}{1 + z^{-1} - z^{-2}}$$

**Solution:**

**a. Direct form I structure:**

$$H(z) = \underbrace{\left[1 + 2z^{-1} - z^{-2}\right]}_{\text{zeros}}\underbrace{\left[\frac{1}{1 + z^{-1} - z^{-2}}\right]}_{\text{poles}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \left[\frac{W(z)}{X(z)}\right]\left[\frac{Y(z)}{W(z)}\right]$$

$$= H_1(z)H_2(z)$$

Where,

$$H_1(z) = \frac{W(z)}{X(z)} = 1 + 2z^{-1} - z^{-2}$$

$$W(z) = X(z) + 2z^{-1}X(z) - z^{-2}X(z)$$

$$w(n) = x(n) + 2x(n-1) - x(n-2)$$

and

$$H_2(z) = \frac{Y(z)}{W(z)} = \frac{1}{1 + z^{-1} - z^{-2}}$$

$$Y(z)\left[1 + z^{-1} - z^{-2}\right] = W(z)$$

$$Y(z) + z^{-1}Y(z) - z^{-2}Y(z) = W(z)$$

$$y(n) = -y(n-1) + y(n-2) + w(n)$$

The direct form I structures is shown in Fig. 4.33 (a).

**b. Direct form II structure:**

$$H(z) = \frac{1 + 2z^{-1} - z^{-2}}{1 + z^{-1} - z^{-2}}$$

$$= \underbrace{\left[\frac{1}{1 + z^{-1} - z^{-2}}\right]}_{\text{poles}}\underbrace{\left[1 + 2z^{-1} - z^{-2}\right]}_{\text{zeros}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \left[\frac{V(z)}{X(z)}\right]\left[\frac{Y(z)}{V(z)}\right]$$

$$= H_1(z)H_2(z)$$

Where,

$$H_1(z) = \frac{V(z)}{X(z)} = \frac{1}{1 + z^{-1} - z^{-2}}$$

$$V(z)\left[1 + z^{-1} - z^{-2}\right] = X(z)$$

$$V(z) + z^{-1}V(z) - z^{-2}V(z) = X(z)$$

$$v(n) = -v(n-1) + v(n-2) + x(n)$$

and $H_2(z) = \frac{Y(z)}{V(z)} = 1 + 2z^{-1} - z^{-2}$

$$Y(z) = V(z) + 2z^{-1}V(z) - z^{-2}V(z)$$

$$y(n) = v(n) + 2v(n-1) - v(n-2)$$

The direct from II structure is shown in Fig. 4.33 (b).

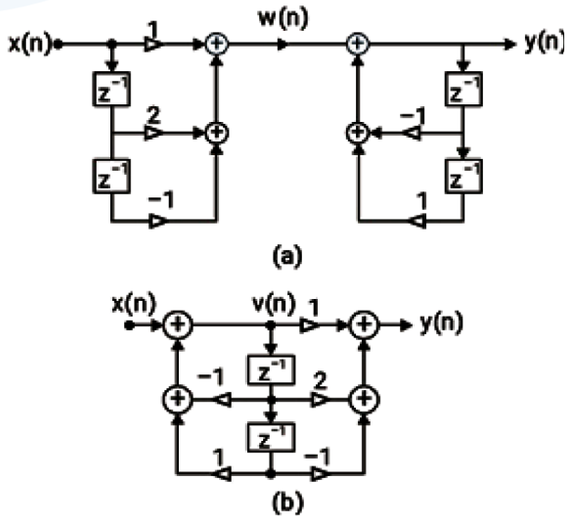



**Fig. 4.33 (a) Direct Form I; (b) Direct Form II**

**12.** An LTI system is described by the difference equation:

$y(n) = a_1y(n-1) + x(n) + b_1(x)(n-1)$

Realise it in direct form I structure and convert it into direct form II structure.

**Solution:**

$y(n) = a_1y(n-1) + x(n) + b_1x(n-1)$

$Y(z) = a_1z^{-1}Y(z) + X(z) + b_1z^{-1}X(z)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1+b_1z^{-1}}{1-a_1z^{-1}}$$

$$= \underbrace{\left[1+b_1z^{-1}\right]}_{\text{zeros}}\underbrace{\left[\frac{1}{1-a_1z^{-1}}\right]}_{\text{poles}}$$

$$= \left[\frac{W(z)}{X(z)}\right]\left[\frac{Y(z)}{W(z)}\right] = H_1(z)H_2(z)$$

Where, $H_1(z) = \dfrac{W(z)}{X(z)} = 1+b_1z^{-1}$

$w(n) = x(n) + b_1x(n-1)$

and $H_2(z) = \dfrac{Y(z)}{W(z)} = \dfrac{1}{1-a_1z^{-1}}$

$y(n) = a_1y(n-1) + w(n)$

The direct form I structure is shown in Fig. 4.34 (a). We can convert direct form I structure into direct form II structure by placing all pole filter $H_2(z)$ before the all zero filter $H_1(z)$. The direct form II structure is shown in Fig. 4.34 (b).

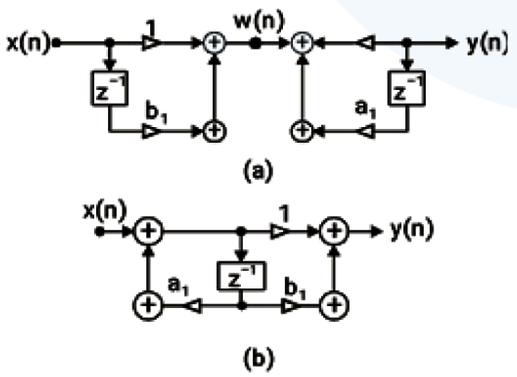



**Fig. 4.34 (a) Direct Form I; (b) Direct Form II**

**13.** Consider a causal LTI system with impulse response h(n) and system function:

$$H(z) = \frac{(1-2z^{-1})(1-4z^{-1})}{z(1-0.5z^{-1})}$$

Draw a direct form II and its signal flow graph for the system and also draw the transposed form of the signal flow graph.

**Solution:**

Consider the given system function.

$$H(z) = \frac{(1-2z^{-1})(1-4z^{-1})}{z(1-0.5z^{-1})} = \frac{z^{-1}\left[1-6z^{-1}+8z^{-2}\right]}{1-0.5z^{-1}}$$

$$= \frac{z^{-1}-6z^{-2}+8z^{-3}}{1-0.5z^{-1}}$$

$$H(z) = \left[\frac{V(z)}{X(z)}\right]\left[\frac{Y(z)}{V(z)}\right] = H_1(z)H_2(z)$$

Where,

$$H_1(z) = \frac{V(z)}{X(z)} = \frac{1}{1-0.5z^{-1}}$$

$v(n) = 0.5v(n-1) + x(n)$

and

$$H_2(z) = \frac{Y(z)}{V(z)} = z^{-1} - 6z^{-2} + 8z^{-3}$$

$y(n) = v(n-1) - 6v(n-2) + 8v(n-3)$

The direct form II structure is shown in Fig. 4.35 (a). Its signal graph is shown in Fig. 4.35 (b) and the transposed form of the signal flow graph is shown in Fig. 4.35 (c)

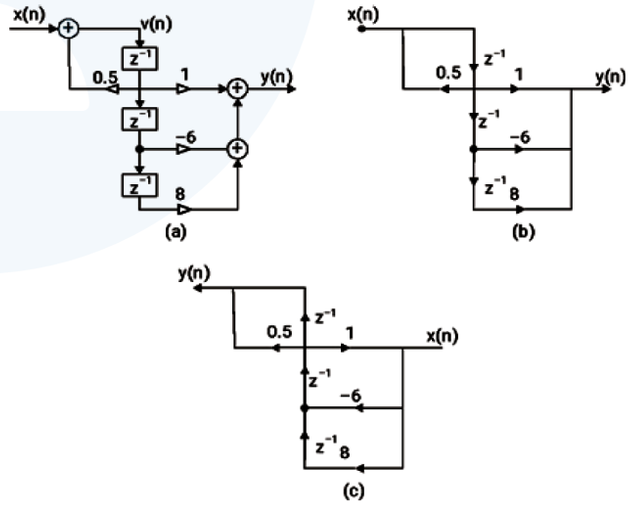



**Fig. 4.35 (a) Direct Form II; (b) Signal Graph; and (c) Form Transposed**

**14.** Obtain the direct form II structure of the system described by difference equation:

$y(n) = 2r\cos(\omega_0)y(n-1) - r^2y(n-2) + x(n)$

$-r\cos(\omega_0)x(n-1)$

**Solution:**

Consider the given difference equation.

$y(n) = 2r\cos(\omega_0)y(n-1) - r^2y(n-2) + x(n) - r\cos(\omega_0)x(n-1)$

$Y(z) = 2r\cos(\omega_0)z^{-1}Y(z) - r^2z^{-2}Y(z) + X(z) - r\cos(\omega_0)z^{-1}X(z)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - r\cos(\omega_0)z^{-1}}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}$$

$$= \underbrace{\left[\frac{1}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}\right]}_{\text{poles}} \underbrace{\left[1 - r\cos(\omega_0)z^{-1}\right]}_{\text{zero s}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \left[\frac{V(z)}{X(z)}\right]\left[\frac{Y(z)}{V(z)}\right]$$

$$= H_1(z)H_2(z)$$

$$H_1(z) = \frac{V(z)}{X(z)} = \frac{1}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}$$

$v(n) = 2r\cos(\omega_0)v(n-1) - r^2v(n-2) + x(n)$

and

$$H_2(z) = \frac{Y(z)}{V(z)} = 1 - r\cos(\omega_0)z^{-1}$$

$y(n) = v(n) - r\cos(\omega_0)v(n-1)$

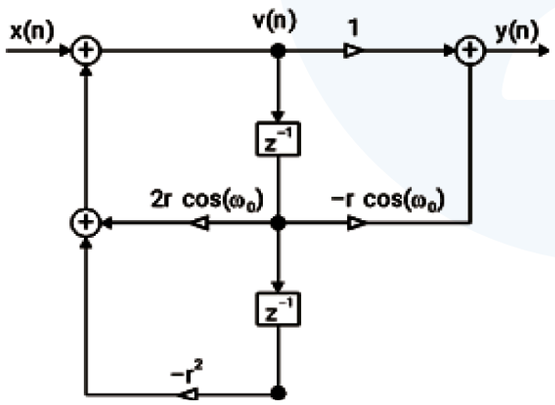



**Fig. 4.36**

**15.** Obtain the cascade and parallel form realization structures for the system function:

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{\left(1 + \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}\right)}$$

**Solution:**

Cascade form structure:

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{\left(1 + \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{2}z^{-1} + \frac{1}{2}z^{-2}\right)}$$

$$= \left[\frac{1 + \frac{1}{4}z^{-1}}{1 + \frac{1}{2}z^{-1}}\right]\left[\frac{1}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}\right]$$

$$= H_1(z)\,H_2(z)$$

Where, $H_2(z)$ is a first order section and $H_1(z)$ is a second order (biquad) section, which are realised using direct from II as shown in Fig. 4.37 (a).

Parallel Form Structure:

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{\left(1 + \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-1}\right)} = \left[\frac{\frac{1}{2}}{1 + \frac{1}{2}z^{-1}}\right] + \left[\frac{\frac{1}{2} - \frac{1}{4}z^{-1}}{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}\right]$$

$$= H_1(z) + H_2(z)$$

Parallel form structure is shown in Fig. 4.37 (b).

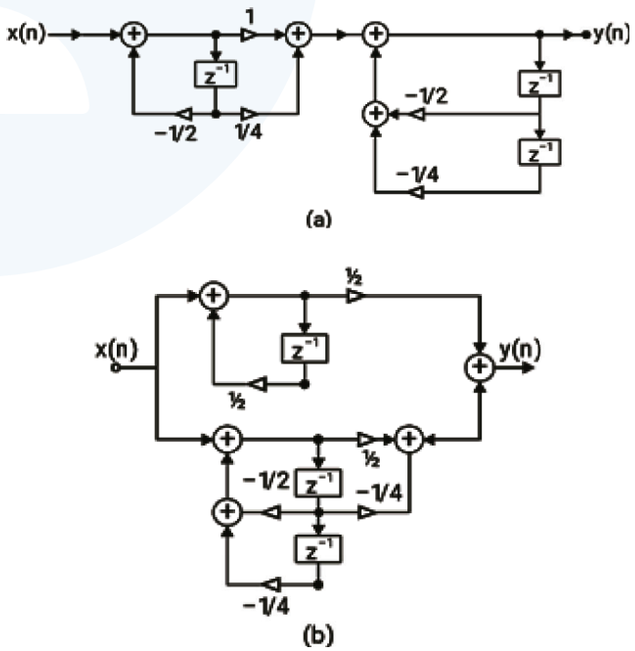



**Fig. 4.37 Structure for Example (a) Cascade Form; (b) Parallel Form**

**16.** Obtain the cascade and parallel form realisations of the given LTI system governed by the difference equation:

$$y(n) = \frac{5}{8}y(n-1) - \frac{1}{16}y(n-2) + x(n)$$
$$-3x(n-1) + 3x(n-2) - x(n-3)$$

**Solution:**

Cascade form structure: The z-transform of this difference equation yields:

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1 - 3z^{-1} + 3z^{-2} - z^{-3}}{1 - \frac{5}{8}z^{-1} + \frac{1}{16}z^{-2}}$$

$$= \frac{\left(1 - z^{-1}\right)^3}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{8}z^{-1}\right)}$$

$$H(z) = [1 - z^{-1}]\left[\frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1}}\right]\left[\frac{1 - z^{-1}}{1 - \frac{1}{8}z^{-1}}\right]$$

$$= H_1(z)H_2(z)H_3(z)$$

which are first order sections.

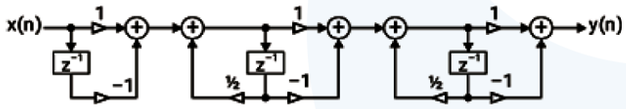



**Fig. 4.38**

**Parallel form structure:** Partial fraction expansion of H(z) yields:

$$H(z) = \left[-112 - 16z^{-1}\right] + \left[\frac{-\frac{4}{3}}{1 - \frac{1}{2}z^{-1}}\right] + \left[\frac{\frac{343}{3}}{1 - \frac{1}{8}z^{-1}}\right]$$

$$= H_z(z) + H_2(z) + H_3(z)$$

Where $H_1(z)$, $H_2(z)$, and $H_3(z)$ are first order sections.

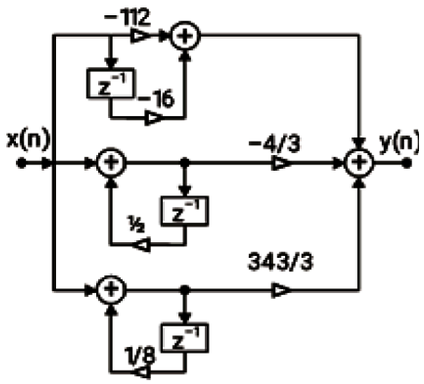



**Fig. 4.39**

## Solved Examples

**17.** Obtain the direct form I and II, cascade and parallel form realization structures for the following system.

$$y(n) = -0.1y(n-1) + 0.72y(n-2) + 0.7x(n) - 0.25x(n-2)$$

**Solution:**

**a. Direct form I structure:** The z-transform of this difference equation yields:

$$\frac{Y(z)}{X(z)} = H(z) = \frac{0.7 - 0.25z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

$$= \underbrace{\left[0.7 - 0.25z^{-2}\right]}_{\text{zeros}}\underbrace{\left[\frac{1}{1 + 0.1z^{-1} - 0.72z^{-2}}\right]}_{\text{poles}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \left[\frac{W(z)}{X(z)}\right]\left[\frac{Y(z)}{W(z)}\right]$$

$$= H_1(z)H_2(z)$$

Where,

$$H_1(z) = \frac{W(z)}{X(z)} = 0.7 - 0.25z^{-2}$$

$w(n) = 0.7x(n) - 0.25x(n-2)$ and

$$H_2 = \frac{Y(z)}{W(z)} = \frac{1}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

$y(n) = -0.1y(n-1) + 0.72y(n-2) + w(n)$

The direct form I structure is shown in Fig. 4.40 (a).

**b. Direct form II structure:** Consider the system function:

$$H(z) = \frac{0.7 - 0.25z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

$$= \underbrace{\left[\frac{1}{1 + 0.1z^{-1} - 0.72z^{-2}}\right]}_{\text{poles}}\underbrace{\left[0.7 - 0.25z^{-2}\right]}_{\text{zeros}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \left[\frac{V(z)}{X(z)}\right]\left[\frac{Y(z)}{V(z)}\right]$$

$$= H_1(z)H_2(z)$$

Where,

$$H_1(z) = \frac{V(z)}{X(z)} = \frac{1}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

$v(n) = -0.1v(n-1) + 0.72v(n-2) + x(n)$

and $$H_2(z) = \frac{Y(z)}{V(z)} = 0.7 - 0.25z^{-2}$$

$y(n) = 0.7v(n) - 0.25v(n-2)$

The direct form II structure is shown in Fig. 4.40 (b).

**c. Cascade form structure:** Consider the system function:

$$H(z) = \frac{0.7 - 0.25z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

$$= \frac{\left(\sqrt{0.7} - 0.5z^{-1}\right)\left(\sqrt{0.7} + 0.5z^{-1}\right)}{\left(1 + 0.9z^{-1}\right)\left(1 - 0.8z^{-1}\right)}$$

$$H(z) = \left[\frac{\sqrt{0.7} - 0.5z^{-1}}{1 + 0.9z^{-1}}\right]\left[\frac{\sqrt{0.7} + 0.5z^{-1}}{1 - 0.8z^{-1}}\right]$$

$$= H_1(z)H_2(z)$$

Where $H_1(z)$ and $H_2(z)$ are first order sections. The cascade form structure is shown in Fig. 4.40 (c).

**d. Parallel form structure:**

$$H(z) = \frac{0.7 - 0.25z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

$$= 0.3472 + \left[\frac{0.2072}{1 + 0.9z^{-1}}\right] + \left[\frac{0.1456}{1 - 0.8z^{-1}}\right]$$

$$= H_1(z) + H_2(z) + H_3(z)$$

Parallel form structure is shown in Fig. 4.40 (d).

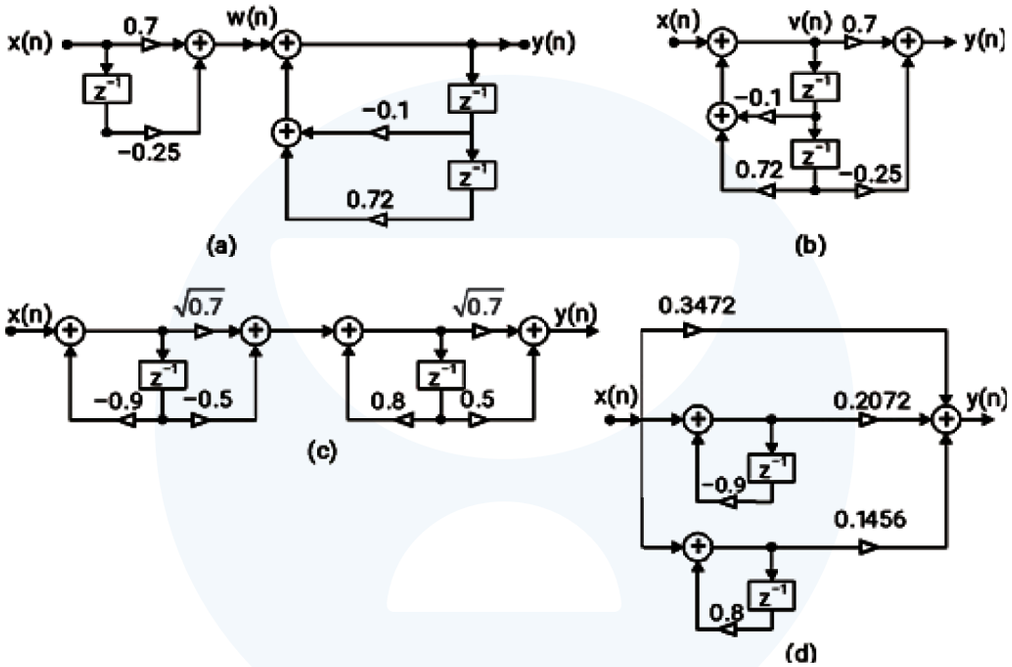



**Fig. 4.40 Structure for Example (a) Direct Form I; (b) Direct Form II; (c) Cascade Form; and (d) Parallel Form**

## Lattice Structure for FIR System

It is used in digital speech processing and in the implementation of adaptive filters.

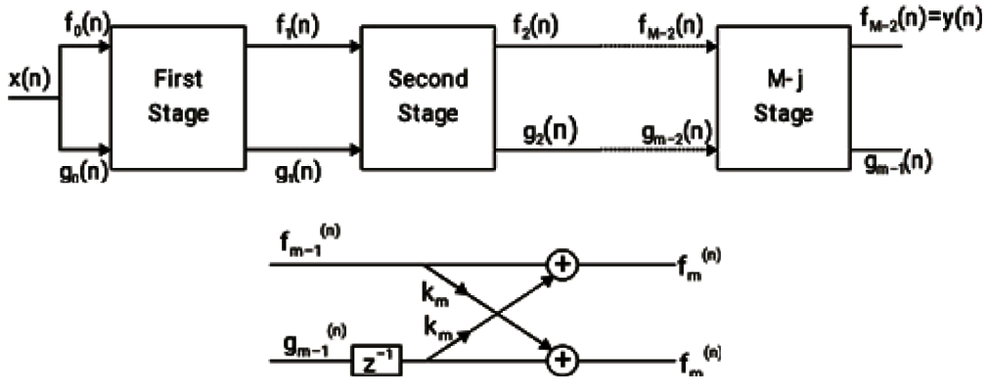



**Fig. 4.41**

### Conversion of Lattice Coefficients to Direct Form Filter Coefficients

The direct form FIR filter coefficients $\{\alpha_m(k)\}$ can be obtained from the lattice coefficients $\{k_i\}$ by using the following relations.

$A_0(z) = B_0(z) = 1$

$A_m(z) = A_{m-1}(z) + k_m z^{-1} B_{m-1}(z),\ m = 1, 2, \ldots M-1$

$B_m(z) = Z^{-m} A_m(z^{-1}),\ m = 1, 2 \ldots M-1$

System Function H(z) of FIR filter is

$$H(z) = \sum_{k=0}^{m} h(k) z^{-k},\ m = 0, 1, 2, M-1$$

H(z) is $m^{th}$ degree polynomial $A_m(z)$

$$H(z) = A_m(z) = \sum_{k=0}^{m} \alpha_m(k) z^{-k} = 1 + \sum_{k=1}^{m} \alpha_m(k) z^{-k},\ m \geq 1$$

$\alpha_m(0) = 1$

$$A_m(z) = 1 + \sum_{k=1}^{m} \alpha_m z^{-k}$$

If x(n) is input sequence of filter $A_m(z)$ and y(n) is the output sequence, we have

$$\boxed{y(n) = x(n) + \sum_{k=1}^{m} \alpha_m(k) x(n-k)}$$

$\alpha_m(0) = 1$

$$\boxed{y(n) = x(n) - \hat{x}(n)}$$

where $\hat{x}(n) = -\sum_{k=1}^{m} \alpha_m(k) x(n-k)$

### Direct Form Realisation of FIR Prediction Filter

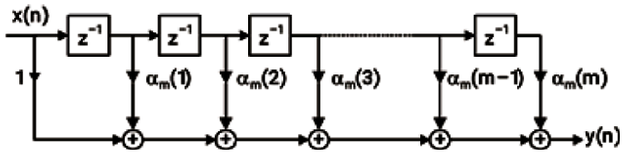



**Fig. 4.42**

## Solved Examples

**18.** Given a three stage lattice filter with coefficients $k_1 = 0.1, k_2 = 0.2$ and $k_3 = 0.3$. Determine the FIR filter coefficients for direct form structure.

**Solution:**

$M - 1 = 3$

$A_0(z) = B_0(z) = 1$

$A_m(z) = A_{m-1}(z) + k_m z^{-1} B_{m-1}(z),\ m = 1, 2, 3 \ldots M-1$

$m = 1, 2, 3$

$B_m(z) = z^{-m} A_m(z^{-1}),\ m = 1, 2, 3, M - 1$

$m = 1, 2, 3$

For m = 1

$A_1(z) = A_0(z) + k_1 z^{-1}$

$= 1 + 0.1 z^{-1}$

$$A_m(z) = 1 + \sum_{k=1}^{m} \alpha_m(k) z^{-k}$$

$A_1(z) = 1 + \alpha_1(1) z^{-1}$

$\boxed{\alpha_1(0) = 1, \alpha_1(1) = 0.1}$

For m = 2

$A_2(z) = A_1(z) + k_2 z^{-1} B_1(z)$

$B_m(z) = z^{-m} A_m(z^{-1})$

$B_1(z) = z^{-1} A_1(z^{-1}) = 0.1 + z^{-1}$

$A_2(z) = A_1(z) + K_2 z^{-1} B_1(z)$

$= \left(1 + 0.1 z^{-1}\right) + 0.2 z^{-1}\left[0.1 + z^{-1}\right]$

$A_2(z) = 1 + 0.12 z^{-1} + 0.2 z^{-2}$

$$A_2(z) = 1 + \sum_{k=1}^{2} \alpha_2(k) z^{-k} = 1 + \alpha_2(1) z^{-1} + \alpha_2(2) z^{-2}$$

$= \alpha_2(0) + \alpha_2(1) z^{-1} \alpha_2(2) z^{-2}$

$\boxed{\alpha_2(0) = 1, \alpha_2(1) = 0.12, \alpha_2(2) = 0.2}$

For m = 3,

$A_3(z) = A_2(z) + k_3 z^{-1} B_2(z)$

$B_2(z) = z^{-2} A_2(z^{-1})$

$= 0.2 + 0.12 z^{-1} + z^{-2}$

$A_3(z) = \left(1 + 0.12 z^{-1} + 0.2 z^{-2}\right) + 0.3 z$

$\left(0.2 + 0.12 z^{-1} + z^{-2}\right)$

$= 1 + 0.18 z^{-1} + 0.236 z^{-2} + 0.3 z^{-3}$

$$A_3(z) = 1 + \sum_{k=1}^{3} \alpha_3(k) z^{-1} = 1 + \alpha_3(1) z^{-1} + \alpha_3(2) z^{-2} + \alpha_3(3) z^{-3}$$

$= \alpha_3(0) + \alpha_3(1) z^{-1} + \alpha_3(2) z^{-2} + \alpha_3(3) z^{-3}$

$\boxed{\alpha_3(0) = 1, \alpha_3(1) = 0.18, \alpha_3(2) = 0.236, \alpha_3(3) = 0.3}$

### Conversion From Direct Form FIR Filter Coefficients to Lattice Coefficients

$$\boxed{A_{m-1}(z) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m^2}}, \quad m = M - 1, M - 2, \ldots \text{ (i)}$$

**Note:**

- $K_m = \alpha_m(m)$
- The above procedure works for as long $|k_m| \neq 1$ as for m = 1, 2, ... M – 1.
- If $|k_m| = 1$ it indicate that $A_{m-1}(z)$ a root on the unit circle. Such a root can be factored out from $A_{m-1}(z)$ and above process is carried out for the reduced order system.

## Solved Examples

**19.** Determine the lattice coefficients corresponding to FIR filter with system functions.

$$H(z) = A_3(z) = 1 + 0.18z^{-1} + 0.236z^{-2} + 0.32z^{-3}$$

**Solution:**

$$A_3(z) = 1 + 0.18z^{-1} + 0.236z^{-2} + 0.3z^{-3}$$

$$= \alpha_3(0) + \alpha_3(1)z^{-1} + \alpha_3(2)z^{-2} + \alpha_3(3)z^{-3}$$

$$\boxed{k_3 = \alpha_3(3) = 0.3}$$

$$B_m(z) = z^{-m}A_m(z^{-1})$$

$$B_3(z) = z^{-3}A_3(z^{-1})$$

$$= 0.3 + 0.236z^{-1} + 0.18z^{-2} + z^{-3}$$

$$A_{m-1}(z) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m^2}, \quad m = M - 1, M - 2, \ldots 1$$

For m = 3, $A_2(z) = \dfrac{A_3(z) - k_3 B_3(z)}{1 - k_3^2}$

$$= 1 + 0.12z^{-1} + 0.2z^{-2}$$

$$= \alpha_2(0) + \alpha_2(1)z^{-1} + \alpha_2(2)z^{-2}$$

$$\boxed{k_2 = \alpha_2(2) = 0.2}$$

$$B_2(z) = z^{-2}A_2(z^{-1}) = 0.2 + 0.02z^{-1} + z^{-2}$$

For m = 2, $A_1(z) = \dfrac{A_2(z) - k_2 B_2(z)}{1 - k_2^2}$

$$= 1 + 0.1z^{-1}$$

$$= \alpha_1(0) + \alpha_1(1)z^{-1}$$

$$\boxed{k_1 = \alpha_1(1) = 0.1}$$

**20.** Realise the following FIR system $H(z) = 1 + 3z^{-1} + 2z^{-2}$ in the following form.

a. Cascade form

b. Lattice form

**Solution:**

a. Cascade form:

$$H(z) = 1 + 3z^{-1} + 2z^{-2}$$

$$= \underbrace{\left(1 + z^{-1}\right)}_{H_1(z)}\underbrace{\left(1 + 2z^{-1}\right)}_{H_2(z)}$$

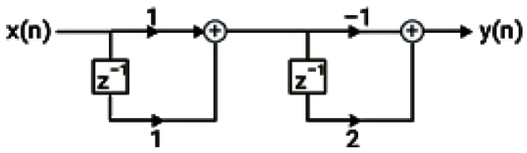



**Fig. 4.43**

b. Lattice form $H(z) = 1 + 3z^{-1} + 2z^{-2}$

$$A_2(z) = \alpha_2(0) + \alpha_2(1)z^{-1} + \alpha_2(2)z^{-2}$$

$$k_2 = \alpha_2(2) = 2$$

$$B_m(z) = z^{-m}A_m(z^{-1})$$

$$B_2(z) = z + 3z^{-1} + z^{-2}$$

$$A_{m-1}(2) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m^2}$$

For m = 2,

$$A_1(z) = \frac{A_2(z) - k_2 B_2(z)}{1 - k_2^2}$$

$$= 1 + z^{-1} = \alpha_1(0) + \alpha_1(1)z^{-1}$$

$$k_1 = \alpha_1(1) = 1$$

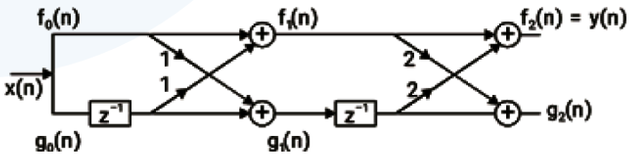



**Fig. 4.44**

$$f_1(n) = f_0(n) + g_0(n-1) = x(n) + x(n-1)$$

$$g_1(n) = f_0(n) + g_0(n-1) = x(n) + x(n-1)$$

$$= f_1(n) + 2g_1(n-1)$$

$$= x(n) + x(n-1) + 2\left[x(n-1) + x(n-2)\right]$$

$$= x(n) + 3x(n-1) + 2x(n-2)$$

$$Y(z) = X(z) + 3z^{-1}X(z) + 2z^{-2}X(z)$$

$$\boxed{H(z) = \frac{Y(2)}{X(z)} = 1 + 3z^{-1} + 2z^{-2}}$$

### Advantages of Lattice Structures

- Lattice structures are modular in nature, i.e., filter order can be increased by adding extra stages and calculating coefficients of the new stages alone.
- Lattice structures are less sensitive to coefficients quantisation effects (finite word length effects) than direct form filter structures.
- Lattice structure are computationally more efficient than other filter structures for the implementation of wavelet transform using filter bank.
- In linear prediction, the lattice structures simultaneously yield the forward and backward prediction errors.
- The FIR filter of class of m requires $\frac{m(m+1)}{2}$ filter coefficients, whereas the lattice form requires any m reflection coefficients $\{k_1\}$. The reason that lattice provides more compact representation for the class of m FIR filter.

## Practice Exercise

### Level I

**1.** The direct form FIR structure is also referred to as a _____ structure.
(A) transposed
(B) cascade
(C) transversal
(D) parallel

**2.** A digital filter is said to be an IIR, if _____.
(A) all poles are outside the unit circle
(B) it oscillates
(C) one of the denominator coefficients is non-zero
(D) the current output depends on previous output

**3.** In high-speed filtering applications _____.
(A) parallel realization is preferred
(B) cascaded realization is preferred
(C) linear realization is preferred
(D) None of the above

**4.** The FIR filter is always stable because all of its poles are _____.
(A) at the origin
(B) at the infinity
(C) at the R.O.C.
(D) None of these

**5.** The canonical form of a structure is _____.
(A) direct-form I
(B) direct-form II
(C) both (A) and (B)
(D) None of the above

**6.** The form of structure that is not used in FIR filters is _____.
(A) direct-form I
(B) direct-form II
(C) cascade
(D) parallel

### Level II

**1.** What is the minimum number of multipliers, adders, and delay elements required to implement a linear-phase filter with h(n) = 0 for n < 0 and n > 15.
(A) 15 multipliers, 15 adders and 15 delay elements.
(B) 8 multipliers, 8 adders, and 15 delay elements.
(C) 8 multipliers, 15 adders, and 15 delay elements.
(D) 15 multipliers, 8 adders, and 15 delay elements.

**2.** What is the frequency response $H(e^{j\omega})$ of the FIR filter that has reflection coefficients $K_1 = K_2 = K_3 = K_4 = 0$ and $K_5 = 1$?
(A) $1 - e^{-j5\omega}$
(B) $e^{-j5\omega}$
(C) $1 + e^{-j5\omega}$
(D) $-e^{-j5\omega}$

**3.** If H(z) is an FIR filter with reflection coefficients $K_1, K_2, \ldots, K_m$, what is the system function of the filter G(z) with reflection coefficients $K_1, 0, K_2, 0, K_3, \ldots, 0, K_m$?

(A) G(z) = H(z)
(B) G(z) = H($z^2$)
(C) G(z) = H($z^{1/2}$)
(D) G(z) = H($z^{-2}$)

**4.** Direct form implementation of the FIR structure as shown in Fig. 1 is also implemented in the Fig. 2 in lattice form. The coefficients $k_1$ and $k_2$ are respectively:

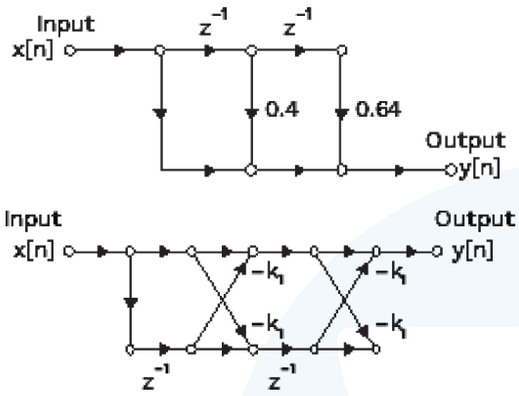



**Fig. 4.45**

(A) 0.640 and −0.244
(B) 0.640 and 0.244
(C) 0.244 and 0.640
(D) −0.244 and −0.640

**5.** A direct form implementation of an LTI system with

$$H(z) = \frac{1}{1 - 0.7z^{-1} + 0.13z^{-2}}$$ is shown in Fig. The values of $a_0$, $a_1$ and $a_2$ are respectively.

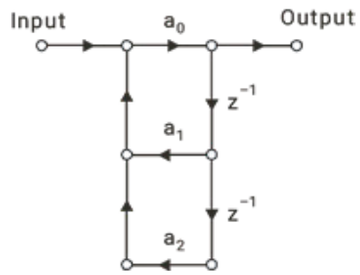



**Fig. 4.46**

(A) 1.0, 0.7 and −0.13
(B) −0.13, 0.7 and 1.0
(C) 1.0, −0.7 and 0.13
(D) 0.13, −0.7 and 1.0

**6.** A discrete time system is shown in the Fig (a). The system function, H(z), of the network is given by:

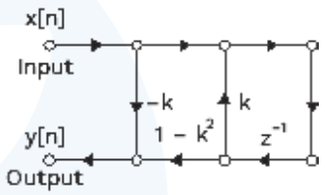



**Fig. 4.47**

(A) $\frac{-k + z^{-1}}{1 - (1 - k^2)z^{-1}}$

(B) $\frac{(1 - k^2) + z^{-1}}{1 - kz^{-1}}$

(C) $\frac{1 - (1 - k^2)z^{-1}}{1 - kz^{-1}}$

(D) $\frac{-k + z^{-1}}{1 - kz^{-1}}$

## Answer Key

**Level I**

**1.** (C) **2.** (D) **3.** (A) **4.** (A) **5.** (B) **6.** (B)

**Level II**

**1.** (C) **2.** (C) **3.** (B) **4.** (D) **5.** (A) **6.** (D)